Data Communication and Networks

Unit -1

1) Draw and explain basic communication system?

The fundamental purpose of a communications system is the exchange of data between two parties.

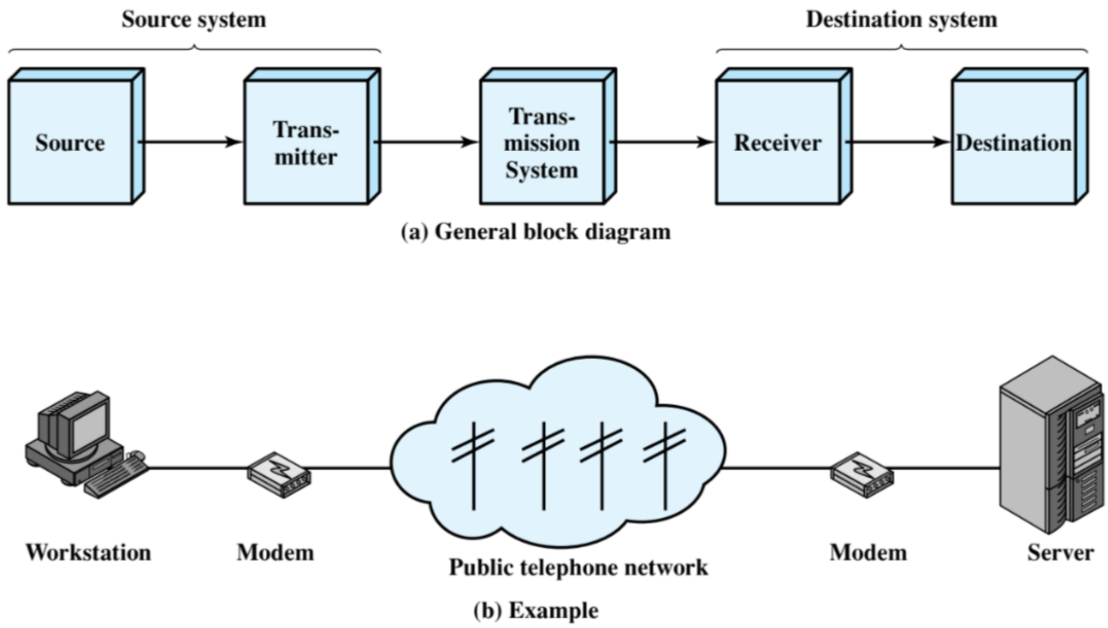


Figure 1.2 Simplified Communications Model

Source - Device that generates data to be transmitted, e.g. telephone, PC

Transmitter - Usually, the data generated by a source system are not transmitted directly in the form in which they were generated. Rather, a transmitter transforms and encodes the information in such a way as to produce electro magnetic signals that can be transmitted across some sort of transmission system. For example, a modem takes a digital bit stream from an attached device such as a personal computer and transforms that bit stream into an analog signal that can be handled by the telephone network.

E.g. Modem takes bits (0's and 1's) and converts into analog signal

Transmission System - Carries data from source to destination

This can be a single transmission line or a complex network connecting source and destination.

Receiver - The receiver accepts the signal from the transmission system and converts it into a form that can be handled by the destination device.

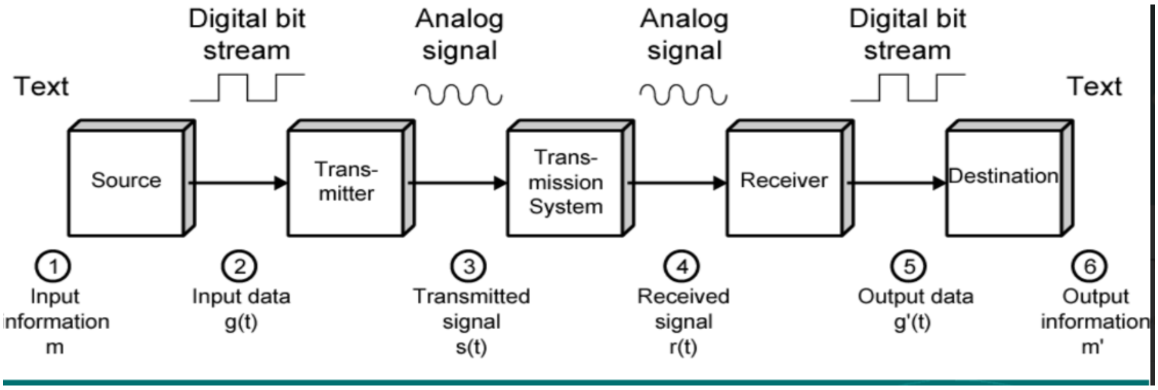
For example, a modem will accept an analog signal coming from a network or transmission line and convert it into a digital bit stream.

Destination - Takes incoming data (dual of source)

2) What is data communication model explain its characteristics.

A network consists of two or more Device that are linked in order to share resources, exchange files, or allow electronic communications.

The computers on a network may be linked through cables, telephone lines, radio waves, satellites, or infrared light beams.



Source: generates data

Example: Terminal, Computer, Phone

Transmitter: transform & encode into EM signal

Example: modem

Transmission system: single Transmission line or

Complex network

Example: Cabling, Microwave, Fibre optics, Radio Frequencies(RF)

Receiver- accepts signal and convert Destination-Takes incoming data.

Example: Printer, Terminal, Computer, Phone.

COMMUNICATION TASKS

A data communications facility is a complex system that cannot create or run itself. Below are the key tasks that must be performed in a data communications system.

- **Transmission system utilization** refers to the need to make efficient use of transmission facilities typically shared among a number of communicating devices.
- To communicate, a device must **interface** with the transmission system
- Once an interface is established, **signal generation** is required for communication The properties of the signal, such as form and intensity, must be such that the signal is
  (1) capable of being propagated through the transmission system, and

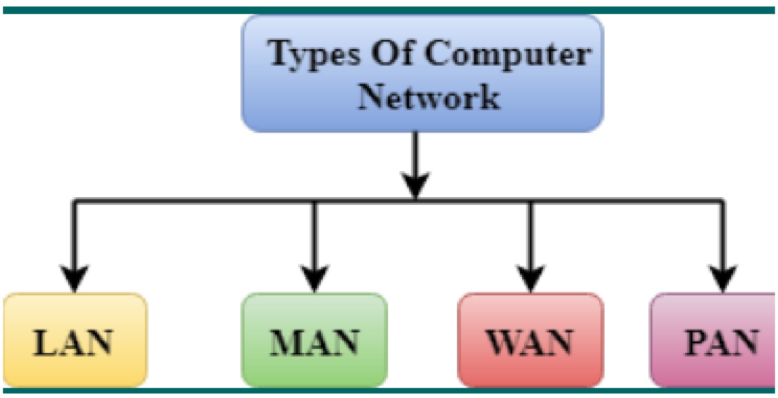
(2) interpretable as data at the receiver.

- There must be **synchronization** between transmitter and receiver, to determine when a signal begins to arrive and when it ends
  there is a variety of requirements for communication between two parties that might be collected under the term **exchange management**. If data are to be exchanged in both directions over a period of time, the two parties must cooperate.
- **Error detection and correction** are required in circumstances where errors cannot be tolerated
- **Flow control** is required to assure that the source does not overwhelm the destination by sending data faster than they can be processed and absorbed
- **addressing and routing**: When more than two devices share a transmission facility, a source system must indicate the identity of the intended destination. The transmission system must assure that the destination system, and only that system, receives the data. Routing is, among the various paths in a network, a specific route through this network must be chosen.
- **Recovery** allows an interrupted transaction to resume activity at the point of interruption or to condition prior to the beginning of the exchange
- **Message formatting** has to do with an agreement between two parties as to the form of the data to be exchanged or transmitted
- **Security** is an important feature in a data communications system. The sender of data may wish to be assured that only the intended receiver actually receives the data. The receiver may wish that the data comes only from the intended sender.
- **Network management** capabilities are needed to configure the system, monitor its status, react to failures and overloads, and plan intelligently for future growth.

3) What is computer network explain different type of computer network.

Computer network can be categorized by their size. A computer network is mainly of four types:

1. LAN(Local Area Network)
2. PAN(Personal Area Network)
3. MAN(Metropolitan Area Network)
4. WAN(Wide Area Network



1. LAN(Local Area Network)

Local Area Network is a group of computers connected to each other in a small area such as building, office.

➢ LAN is used for connecting two or more personal computers through a communication medium such as twisted pair, coaxial cable, etc.

➢ It Is less costly as it is built with inexpensive hardware such as hubs, network adapters, and Ethernet cables.

➢ the data is transferred at an extremely faster rate in Local Area Network.

➢ Local Area Network provides higher security

2. PAN(Personal Area Network)

- Personal Area Network is a network arranged within an individual person, typically within a range of 10 meters.
- Personal Area Network is used for connecting the computer devices of personal use is known as Personal Area Network.
- Thomas Zimmerman was the first research scientist to bring the idea of the Personal Area Network. λ Personal Area Network covers an area of 30 feet.
- Personal area network are the Laptop, mobile phones, media player and play stations.

There are two types of Personal Area Network:

1. Wireless Personal Area Network: Wireless Personal Area Network is developed by simply using wireless technologies such as Wi-Fi, Bluetooth.
   It is a low range network.

2.Wired Personal Area Network: Wired Personal Area Network is created by using the USB.

Examples Of Personal Area Network:

- Body Area Network: Body Area Network is a network that moves with a person. For example, a mobile network moves with a person. Suppose a person establishes a network connection and then creates a connection with another device to share the information.

- Offline Network: An offline network can be created inside the home, so it is also known as a home network. A home network is designed to integrate the devices such as printers, computer, television but they are not connected to the internet.
- Small Home Office: It is used to connect a variety of devices to the internet and to corporate network using a VPN

3. **MAN(Metropolitan Area Network)**

- ➢ A metropolitan area network is a network that covers a larger geographic area by interconnecting a different LAN to form a larger network.
- ➢ Government agencies use MAN to connect to the citizens and private industries.
- ➢ In MAN, various LANs are connected to each other through a telephone exchange line.
- ➢ The most widely used protocols in MAN are RS-232, Frame Relay, ATM, ISDN, OC-3, ADSL, etc.

It has a higher range than Local Area Network(LAN).

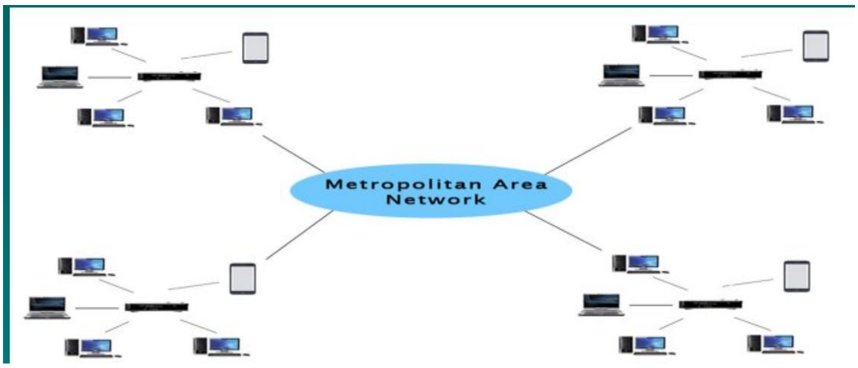
Uses Of Metropolitan Area Network:

MAN is used in communication between the banks in a city.

It can be used in an Airline Reservation.

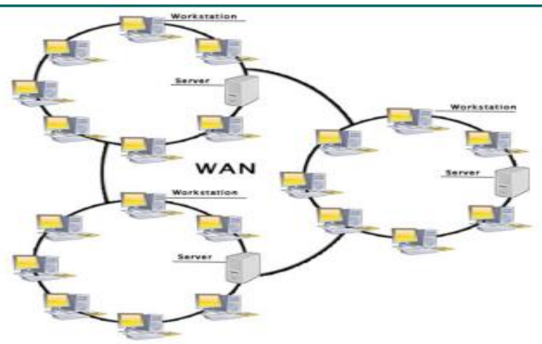
It can be used in a college within a city.

It can also be used for communication in the military.



4. **WAN(Wide Area Network)**

- A Wide Area Network is a network that extends over a large geographical area such as states or countries.
- A Wide Area Network is quite bigger network than the LAN.
- A Wide Area Network is not limited to a single location, but it spans over a large geographical area through a telephone line, fibre optic cable or satellite links.
- The internet is one of the biggest WAN in the world.
- Examples of Wide Area Network: Mobile Broadband.



Advantages of the Wide Area Network:

- Geographical area: A Wide Area Network provides a large geographical area.
- Centralized data: In case of WAN network, data is centralized. Therefore, we do not need to buy the emails, files or back up servers.
- Get updated files: Software companies work on the live server. Therefore, the programmers get the updated files within seconds.
- Exchange messages: In a WAN network, messages are transmitted fast. Eg-Facebook, Whatsapp
- Sharing of software and resources: In WAN network, we can share the software and other resources like a hard drive, RAM.
- Global business: We can do the business over the internet globally.
- High bandwidth: If we use the leased lines for our company then this gives the high bandwidth.

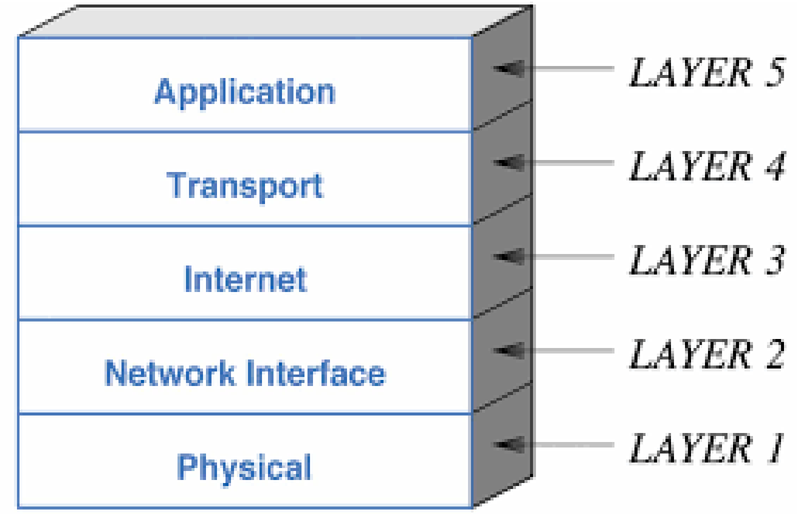
Disadvantages of Wide Area Network:

- Security issue: A WAN network has more security issues as compared to LAN and MAN network as all the technologies are combined together that creates the security problem.
- Needs Firewall & antivirus software: The data is transferred on the internet which can be changed or hacked by the hackers, so the firewall needs to be used. Some people can inject the virus in our system so antivirus is needed to protect from such a virus.
- High Setup cost: An installation cost of the WAN network is high as it involves the purchasing of routers, switches.

- Troubleshooting problems: It covers a large area so fixing the problem is difficult.

4) Explain the working of TCP/IP model?

- Developed by US Defense Advanced Research Project Agency (DARPA).
- ARPANET is packet switched network used by the global Internet.
- protocol suite is a large collection of standardized protocols.
- The TCP/IP model consists of five layers: the application layer, transport layer, internet layer, network access layer and physical layer.



1. Application Layer

- An application layer is the topmost layer.
- This layer allows the user to interact with the application.
- When one application layer protocol wants to communicate with another application layer, it forwards its data to the transport layer.
- It forwards its data to the transport layer.
- Types of apllication like HTTP, SMTP, FTP etc.

2. Transport Layer (TCP)

- Called as Host to Host layer.
- Common layer shared by all applications.
- Responsible for the reliability, flow control, and correction of data which is being sent over the network.

- Data is received in same order as sent.
- TCP, UDP.
- Segmentation -- Segment.
- Flow control, Connection oriented &connection less transfer.

3. Internet Layer (IP)

- Concerned with routing data across.
- multiple interconnected networks using IP protocol.
- Implemented in end systems and routers.
- Routers connect two networks and relays data between them.
- TCP segment + control info ->IP Datagram.
- Router.

4. Network Access Layer

- Exchange of data between 2 end systems in a same network.
- The functions carried out by this layer are encapsulating the IP datagram into frames transmitted by the network and mapping of IP addresses into physical addresses.The protocols used by this layer are ethernet, token ring, FDDI, X.25, frame relay.
- Concerned with issues like : destination address is required
- Sender can invoke services like priority
- NAP N/w Access Protocol
- Packet/Frame

5. Physical Layer

- Concerned with physical interface between computer and network.
- Concerned with issues like:

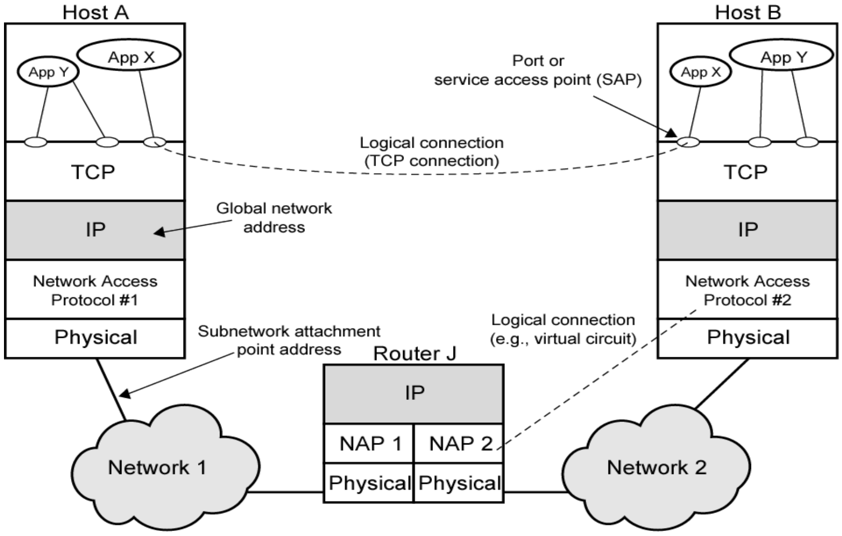
Characteristics of transmission medium.

Signal levels.

Nature of signal.

Data rates.

Other related matters.

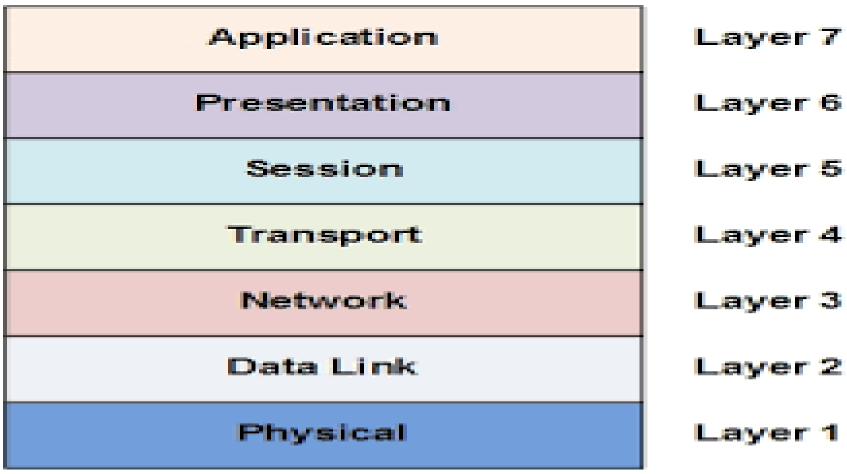


Transmission Control Protocol (TCP)

- Connection oriented service usual transport layer is (TCP)
- provides a reliable connection
- TCP segment - basic unit
- TCP tracks segments between entities for each connection
-

5) Explain the function OSI model.

The Open Systems Interconnection (OSI) model consists of seven layers:



Each layer having specific functionality to perform. All these 7 layers work collaboratively to transmit the data from one person to another.

The functions of the physical layer are:

- Bit synchronization: The physical layer provides the synchronization of the bits by providing a clock. This clock controls both sender and receiver thus providing synchronization at bit level.

- Bit rate control: The Physical layer also defines the transmission rate i.e. the number of bits sent per second.
- Physical topologies: Physical layer specifies the way in which the different, devices/nodes are arranged in a network i.e. bus, star or mesh topology.
- Transmission mode: Physical layer also defines the way in which the data flows between the two connected devices. The various transmission modes possible are: Simplex, half-duplex and full-duplex

The functions of the data Link layer are:

- Framing: Framing is a function of the data link layer. It provides a way for a sender to transmit a set of bits that are meaningful to the receiver. This can be accomplished by attaching special bit patterns to the beginning and end of the frame.
- Physical addressing: After creating frames, Data link layer adds physical addresses (MAC address) of sender and/or receiver in the header of each frame.
- Error control: Data link layer provides the mechanism of error control in which it detects and retransmits damaged or lost frames.
- Flow Control: The data rate must be constant on both sides else the data may get corrupted thus, flow control coordinates that amount of data that can be sent before receiving acknowledgement.
- Access control: When a single communication channel is shared by multiple devices, MAC sub-layer of data link layer helps to determine which device has control over the channel at a given time.

The functions of the Network layer are:

- Routing: The network layer protocols determine which route is suitable from source to destination. This function of network layer is known as routing.
- Logical Addressing: In order to identify each device on internetwork uniquely, network layer defines an addressing scheme. The sender & receiver's IP address are placed in the header by network layer. Such an address distinguishes each device uniquely and universally.

The functions of the transport layer are:

- Segmentation and Reassembly: This layer accepts the message from the (session) layer, breaks the message into smaller units. Each of the segment produced has a header associated with it. The transport layer at the destination station reassembles the message.
- Service Point Addressing: In order to deliver the message to correct process, transport layer header includes a type of address called service point address or port address. Thus by specifying this address, transport layer makes sure that the message is delivered to the correct process.

The services provided by the transport layer:

● Connection Oriented Service
● Connection less service

The functions of the session layer are :

- Session establishment, maintenance and termination: The layer allows the two processes to establish, use and terminate a connection.
- Synchronization : This layer allows a process to add checkpoints which are considered as synchronization points into the data. These synchronization point help to identify the error so that the data is re-synchronized properly, and ends of the messages are not cut prematurely and data loss is avoided.
- Dialog Controller : The session layer allows two systems to start communication with each other in half-duplex or full-duplex.

The functions of the presentation layer are :

- Encryption/ Decryption : Data encryption translates the data into another form or code. The encrypted data is known as the cipher text and the decrypted data is known as plain text. A key value is used for encrypting as well as decrypting data.
- Compression: Reduces the number of bits that need to be transmitted on the network.

The functions of the Application Layer Are:

- These applications produce the data, which has to be transferred over the network.
- This layer also serves as a window for the application services to access the network and for displaying the received information to the user

6) Differentiate between synchronous and asynchronous transmission.

Synchronous transmission	Asynchronous transmission
Data is sent in form of blocks or frames.	Data is sent in form of byte or character.
Transmission is the full duplex type	Transmission is the half duplex type
Between sender and receiver the synchronization is compulsory.	It does not require synchronization.
Transmission speed is fast	Transmission speed is slower
Cost is expensive	Cost is economical
There are gaps between the data	There is no gap present between data.

Time interval is constant	Time interval is random
Used in Chat Rooms, Telephonic Conversations, Video conferencing	Used in Email, Forums, Letters

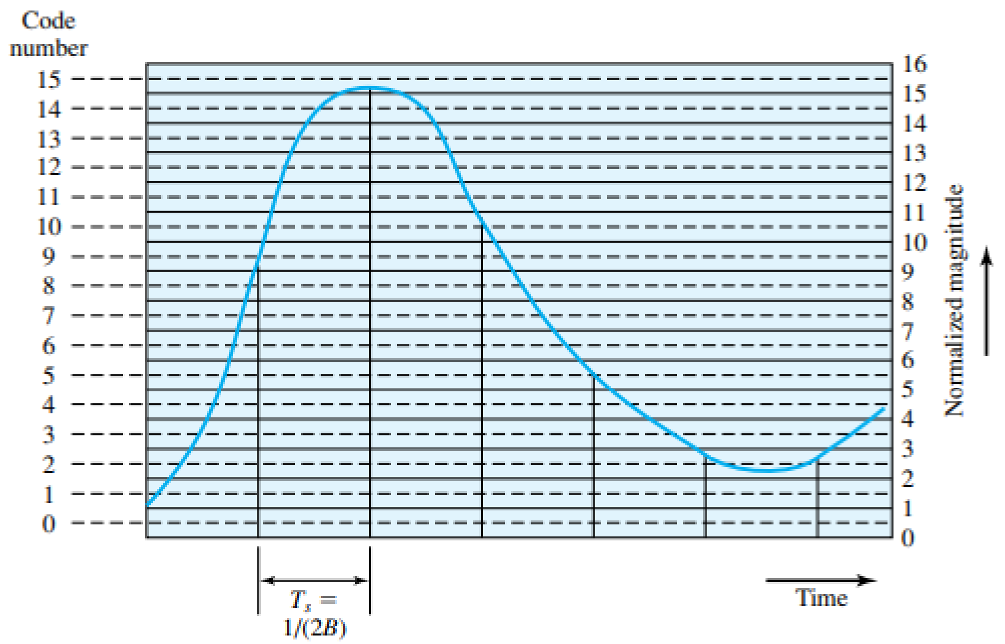
7) Describe the PCM pulse code modulation technique. [Explain PCM with the non-linear encoding]

It is a digital representation of an Analog signal where magnitude of the signal is sampled regularly at uniform intervals then quantized to a series symbols in a numeric code.

In this Analog data should be converted to digital form.

The Analog samples are called pulse amplitude modulation (PAM) samples. To convert to digital, each of these Analog samples must be assigned a binary code.

Eg: Consider a signal for a duration of 2 second in analog signal values are continuous



PAM value	1.1	9.2	15.2	10.8	5.6	2.8	2.7
Quantized code number	1	9	15	10	5	2	2
PCM code	0001	1001	1111	1010	0101	0010	0010

Thus, PCM starts with a continuous-time, continuous-amplitude (analog) signal, from which a digital signal (Figure 5.17).

The digital signal consists of blocks of n bits, where each n-bit number is the amplitude of a PCM pulse. On reception, the process is reversed to reproduce the analog signal. Notice, however, that this process violates the terms of the sampling theorem. By quantizing the PAM pulse, the original signal is now only approximated and cannot be recovered exactly. This effect is known as quantizing error or Quantizing.

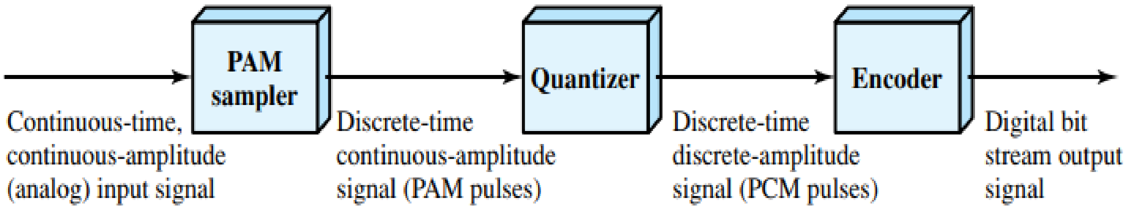


Figure 5.17 PCM Block Diagram

noise. The signal-to-noise ratio for quantizing noise can be expressed as [GIBS93]

SNRdB = 20 log 2n + 1.76 dB = 6.02n + 1.76 dB

Thus each additional bit used for quantizing increases SNR by about 6 dB, which is a factor of 4.

Typically, the PCM scheme is refined using a technique known as nonlinear encoding, which means, in effect, that the quantization levels are not equally spaced. The problem with equal spacing is that the mean absolute error for each sample is the same, regardless of signal level. Consequently, lower amplitude values are relatively more distorted. By using a greater number of quantizing steps for signals of low amplitude, and a smaller number of quantizing steps for signals of large amplitude, a marked reduction in overall signal distortion is achieved (e.g., see Figure 5.18).

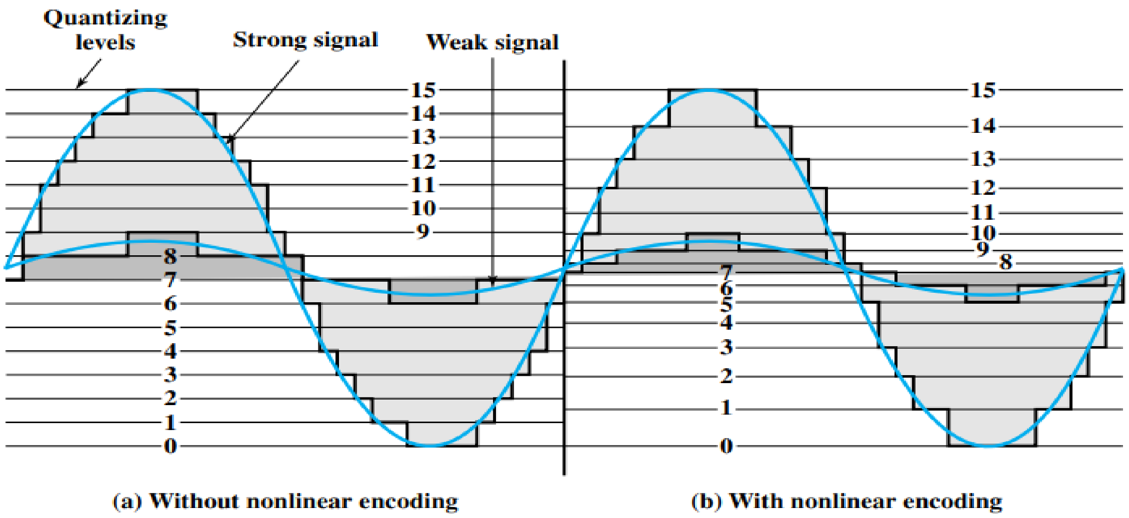


Figure 5.18 Effect of Nonlinear Coding

8) Elaborate parity checker error detection with an example.

- The simplest error-detecting scheme is to append a parity bit to the end of a block of data.
- A typical example is character transmission, in which a parity bit is attached to each 7-bit IRA character.
- The value of this bit is selected so that the character has an even number of 1s (even parity) or an odd number of 1s (odd parity).
- Note, however, that if two (or any even number) of bits are inverted due to error, an undetected error occurs. Typically, even parity is used for synchronous transmission and odd parity for asynchronous transmission.
- The use of the parity bit is not foolproof, as noise impulses are often long enough to destroy more than one bit, particularly at high data rates.

EXAMPLE :If the transmitter is transmitting an IRA character G (1110001) and using odd parity, it will append a 1 and transmit 11110001.2 The receiver examines the received character and, if the total number of 1s is odd, assumes that no error has occurred. If one bit (or any odd number of bits) is erroneously inverted during transmission (for example, 11100001), then the receiver will detect an error.

9) What is CRC? Given P=110101 and M=1010001101 find FCS using

i) Using modulo 2 arithmetic method

ii) Using polynomial method

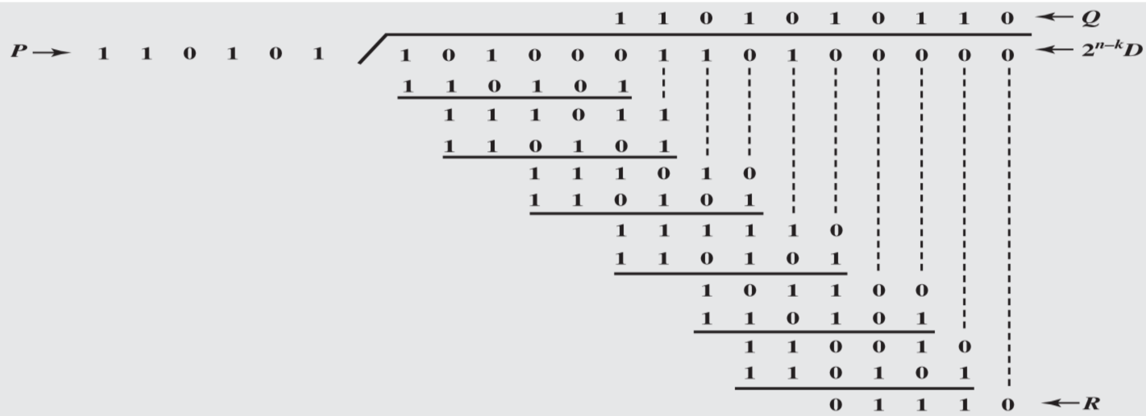
- One of the most common, and one of the most powerful, error-detecting codes is the cyclic redundancy check (CRC), which can be described as follows. Given a k-bit block of bits, or message, the transmitter generates an sequence, known as a frame check sequence (FCS), such that the resulting frame, consisting of n bits, is exactly divisible by some predetermined number. The receiver then divides the incoming frame by that number and, if there is no remainder, assumes there was no error.

Using modulo -2 arithmetic method

1. Given Thus, and

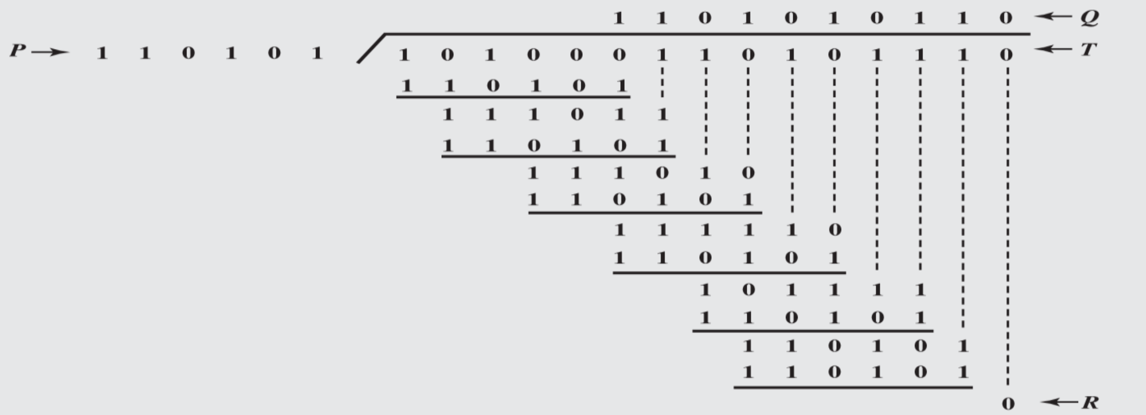
2. The message is multiplied by yielding 101000110100000.

3. This product is divided by P: 25 , n = 15, k = 10, 1n - k2 = 5. FCS R = to be calculated 15 bits2 Pattern P = 110101 16 bits2 Message M = 1010001101 110 bits



4. The remainder is added to 2^5D to give T = 101000110101110, which is transmitted.

5. If there are no errors, the receiver receives T intact. The received frame is divided by P:



Because there is no remainder, it is assumed that there have been no errors.

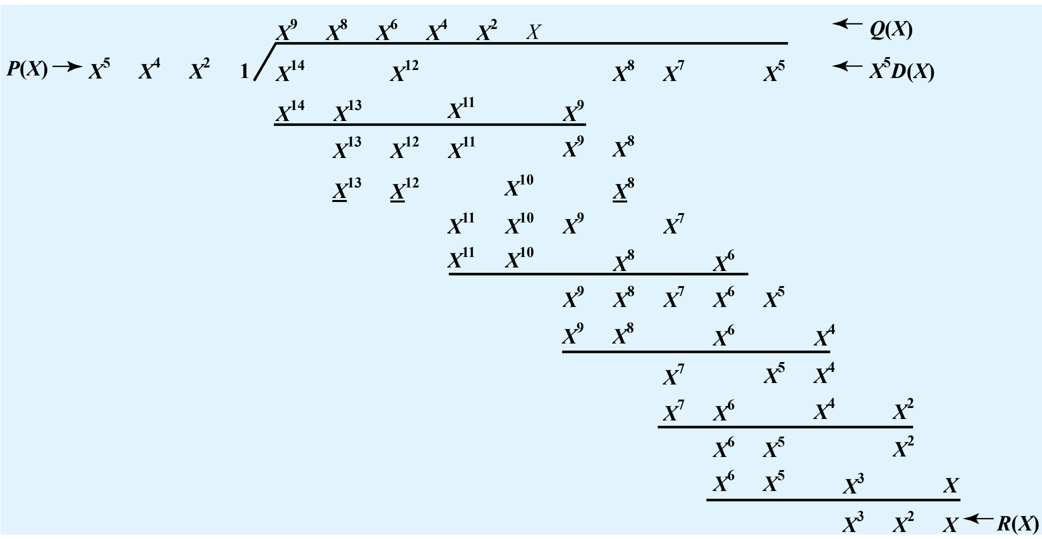
Using polynomial method

Using the preceding example,

for M = 1010001101, we have M= $x^9 + x^7 + x^3 + x^2 + 1$,

and for P = 110101, we have P = $x^5 + x^4 + x^2 + 1$.

S=$x^{14}+x^{12}+x^8+x^7+x^5$



Transmit T to Receiver T=S+R

$T = x^{14} + x^{12} + x^8 + x^7 + x^5 + x^3 + x^2 + x$

T/p=>

$$
\begin{array}{r|l}
x^5 + x^4 + x^2 + 1 & x^{14} + x^{12} + x^8 + x^7 + x^5 + x^3 + x^2 + x \\
 & x^{14} + x^{13} + x^{11} + x^9 \\ \hline
 & x^{13} + x^{12} + x^{11} + x^9 + x^8 \\
 & x^{13} + x^{12} + x^{10} + x^8 \\ \hline
 & x^{11} + x^{10} + x^9 + x^7 \\
 & x^{11} + x^{10} + x^8 + x^6 \\ \hline
 & x^9 + x^8 + x^7 + x^6 + x^5 \\
 & x^9 + x^8 + x^6 + x^4 \\ \hline
 & x^7 + x^5 + x^4 + x^3 + x^2 \\
 & x^7 + x^6 + x^4 + x^2 \\ \hline
 & x^6 + x^5 + x^3 + x \\
 & x^6 + x^5 + x^3 + x \\ \hline
 & 0
\end{array}
$$

10) a. Encode following data 1101011011 using

i) Bipolar AMI

ii) Manchester

iii)Differential Manchester

iv)NRZ -I

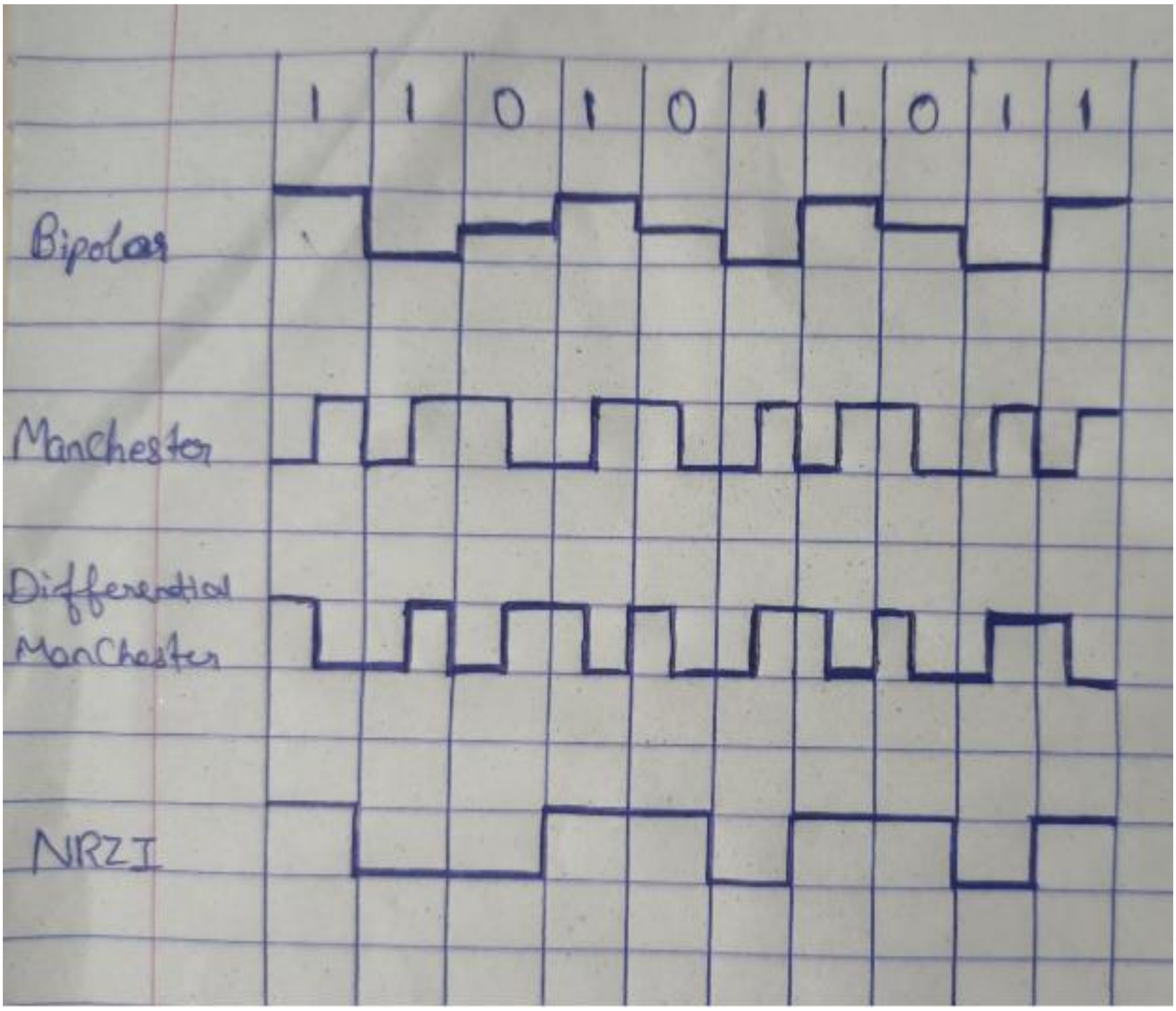
Explain all 4 techniques

b. Construct the data using modulation technique Using

i) ASK ii)FSK iii)PSK

Explain all 3 technique.

a)



i) **Bipolar AMI:**

->Zero –- no line signal

->One –-- +/- level alternating for successive ones

->One –--pulses alternate in polarity

Bipolar Alternate Mark Inversion encoding represents 1 bits with alternating positive and negative voltages, while 0 bits are represented by no signal. This technique reduces DC bias and aids synchronization.

Advantage: no loss of sync if a long string of ones occur and easy error detection

ii) **Manchester:**

->has transition in middle of each bit period

-> transition serves as clock and data

low to high represents-----1

high to low represents -----0

Manchester encoding uses transitions to represent data, with a transition from low to high indicating a 0 bit, and a transition from high to low representing a 1 bit. It provides self-clocking and guarantees transitions for synchronization.

iii) **Differential Manchester:**

->Mid bit transition is clocking only

transition at start of bit period representing--- 0

no transition at start of bit period representing—1

Differential Manchester encoding ensures a transition at the middle of each bit, regardless of the value. A transition at the beginning represents a 0, while no transition at the beginning indicates a 1. It provides self-clocking and eliminates the need for a separate clock signal.

iv) **NRZ-I:**

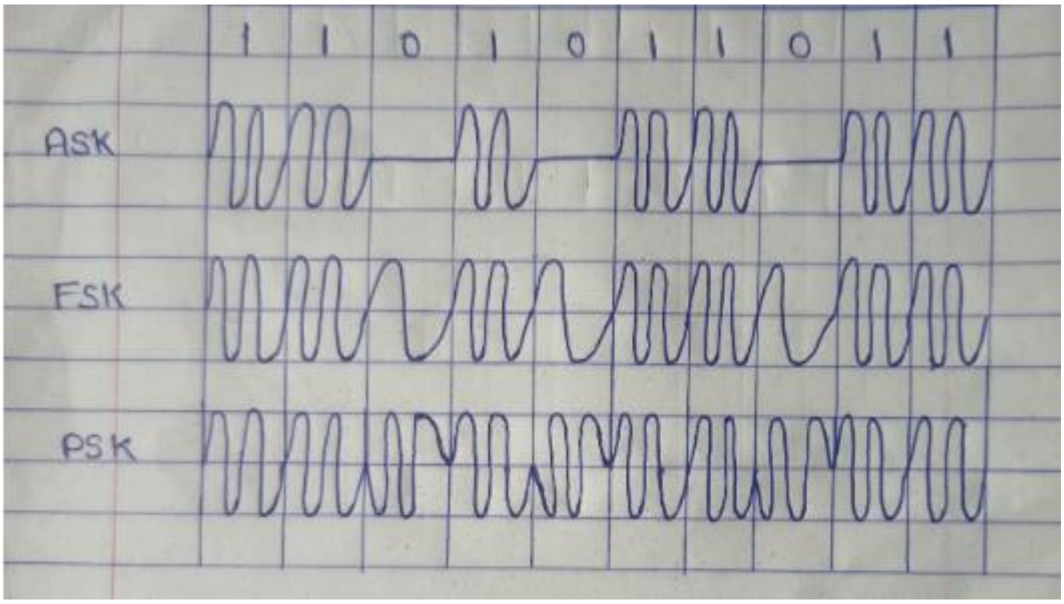
->Maintains constant voltage pulse for duration of bit time

0--- No transition

1---- Transition at the beginning of interval

Non-Return-to-Zero Inverted encoding represents a 1 bit as a polarized signal and inverts the polarity for each 0 bit. It ensures a transition in the middle of each bit, but consecutive 1 bits can cause synchronization issues.

b)



i) **ASK** (Amplitude Shift Keying): ASK is a digital modulation technique where the amplitude of a carrier signal is varied to represent digital data. The amplitude is changed between two levels, typically high and low, to represent 0s and 1s.

ii**)** **FSK** (Frequency Shift Keying): FSK is a modulation technique where the frequency of a carrier signal is shifted between two distinct frequencies to represent digital data. Each frequency represents a different binary value.

iii) **PSK** (Phase Shift Keying): PSK is a modulation technique where the phase of a carrier signal is shifted to represent digital data. The phase is changed between specific angles, typically 0° and 180°, to represent 0s and 1s.

11) Differentiate between OSI and TCP/IP model

OSI Model	**TCP/IP Model**
OSI stands for Open Systems Interconnection.	TCP/IP stands for Transmission Control Protocol/Internet Protocol
OSI model has been developed by ISO (International Standard Organization).	It was developed by ARPANET (Advanced Research Project Agency Network).
It has 7 layers.	It has 4 layers.
It is an independent standard and generic protocol used as a communication gateway between the network and the end user.	It consists of standard protocols that lead to the development of an internet. It is a communication protocol that provides the connection among the hosts.
In this model, the session and presentation layers are separated, i.e., both the layers are different.	In this model, the session and presentation layer are not different layers. Both layers are included in the application layer.
In the OSI model, the transport layer provides a guarantee for the delivery of the packets.	The transport layer does not provide the surety for the delivery of packets.

	But still, we can say that it is a reliable model.
Protocols in the OSI model are hidden and can be easily replaced when the technology changes.	In this model, the protocol cannot be easily replaced.
OSI model defines the services, protocols, and interfaces as well as provides a proper distinction between them. It is protocol independent.	In the TCP/IP model, services, protocols, and interfaces are not properly separated. It is protocol dependent.
It is low in usage.	It is mostly used.

12) <u>What is design goal of scrambling technique. Explain 2 scrambling technique with example</u>

The purpose of a computer network is to transfer data from one location to another. We can send both digital and analog data. Additionally, data transmissions might be either digital or analog. As a result, in order to transmit data using signals, we must be able to convert the data into the appropriate signals. This conversion can be done from analog to analog, analog to digital, digital to digital, or digital to digital.

Scrambling is a method that offers synchronization without increasing the number of bits. Continuous sequences of zeros in techniques like Bipolar AMI cause synchronization issues. Scrambling is one approach to this problem. It is a technique that does not increase the number of bits and does provide synchronization.

The idea behind this approach is simple: Sequences that would result in a constant voltage level on the line are replaced by filling sequences that will provide sufficient transitions for the receiver's clock to maintain synchronization. The filling sequence must be recognized by the receiver and replaced with the original data sequence. The filling sequence is the same length as the original sequence, so there is no data rate penalty. The design goals for this approach can be summarized as follows:

• No dc component

• No long sequences of zero-level line signals

• No reduction in data rate

• Error-detection capability

There are two common scrambling techniques:

1. B8ZS(Bipolar with 8-zero substitution)
2. HDB3(High-density bipolar3-zero)

B8ZS(Bipolar with 8-zero substitution): This technique is similar to Bipolar AMI except when eight consecutive zero-level voltages are encountered they are replaced by the sequence, "000VB0VB".

Here,V(Violation), is a non-zero voltage which means the signal has the same polarity as the previous non-zero voltage. Thus it is a violation of the general AMI technique and B(Bipolar), is also a non-zero voltage level that is in accordance with the AMI rule (i.e., opposite polarity from the previous non-zero voltage).

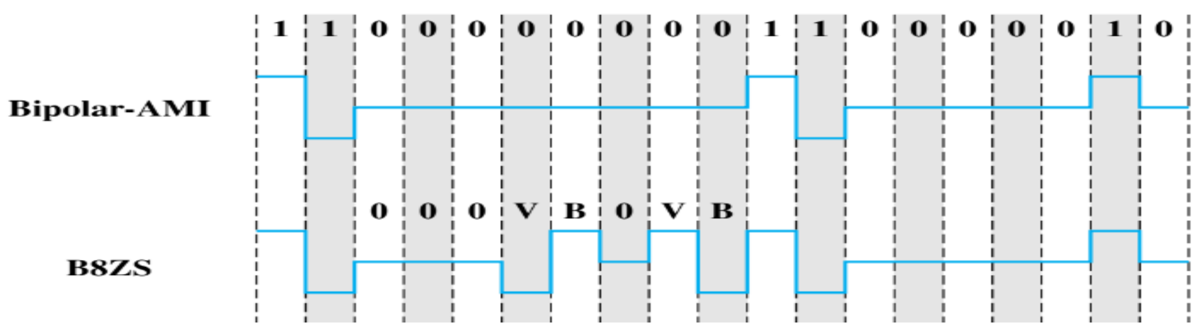
Rules:

• If an octet of all zeros occurs and the last voltage pulse preceding this octet was positive, then the eight zeros of the octet are encoded as 000+ -0- +.

• If an octet of all zeros occurs and the last voltage pulse preceding this octet was negative, then the eight zeros of the octet are encoded as 000- +0+ -.

This coding scheme is widely used in North America.

Example:



HDB3(High-density bipolar3-zero): In this technique, four consecutive zero-level voltages are replaced with a sequence "000V" or "B00V". Rules for using these sequences:

- If the number of nonzero pulses after the last substitution is odd, the substitution pattern will be "000V", this helps in maintaining a total number of nonzero pulses even.
- If the number of nonzero pulses after the last substitution is even, the substitution pattern will be "B00V". Hence even the number of nonzero pulses is maintained again.

This coding scheme is widely used in Europe and Japan.

Rules:

	Number of Bipolar Pulses (ones) since Last Substitution	
Polarity of Preceding Pulse	**Odd**	**Even**
-	000-	+00+
+	000+	-00-

Example: